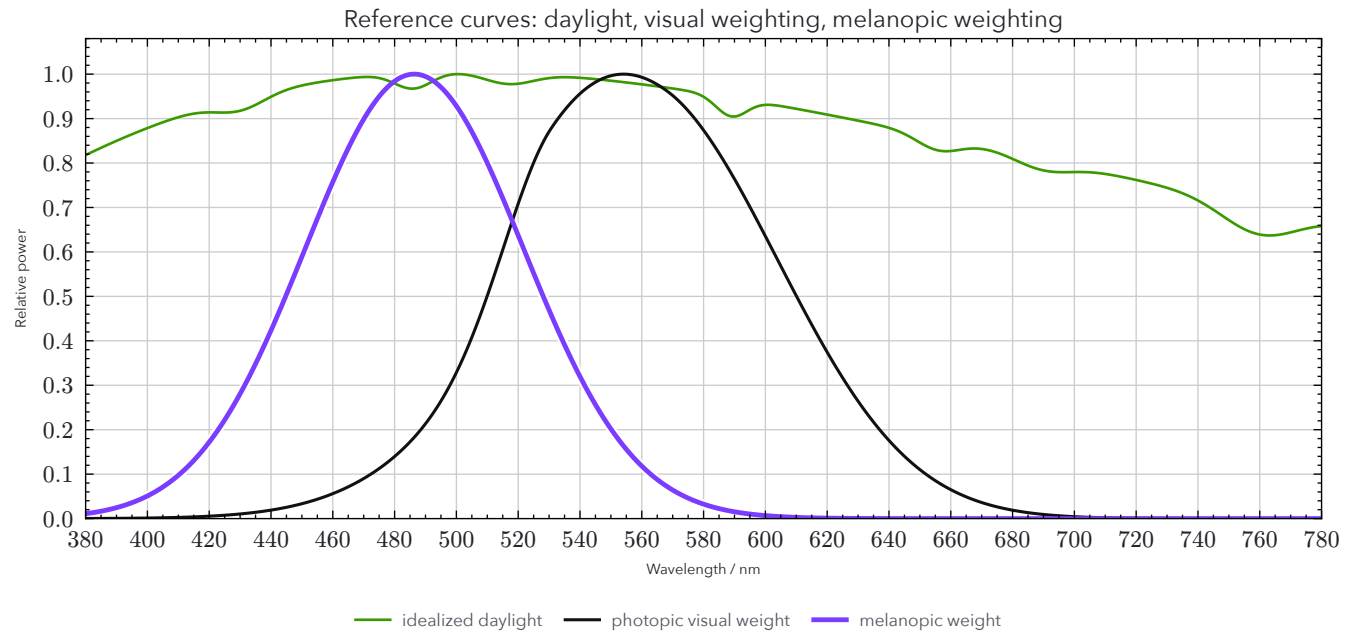


Spectrum is read by more than one system.

A lamp can look visually adequate while landing very differently on the biological sensitivity curve.

Photopic vision and melanopic response do not weight wavelengths the same way. The design problem starts here: "white" is a visual appearance, not a circadian specification.



VISUAL LAYER

Photopic weighting

This is the conventional visual brightness layer. It explains much of what looks bright to the eye.

CLOCK LAYER, FILL: VIOLET

Melanopic weighting

This curve is centered in the blue-cyan region. It is easy to hide in prose and essential in design.

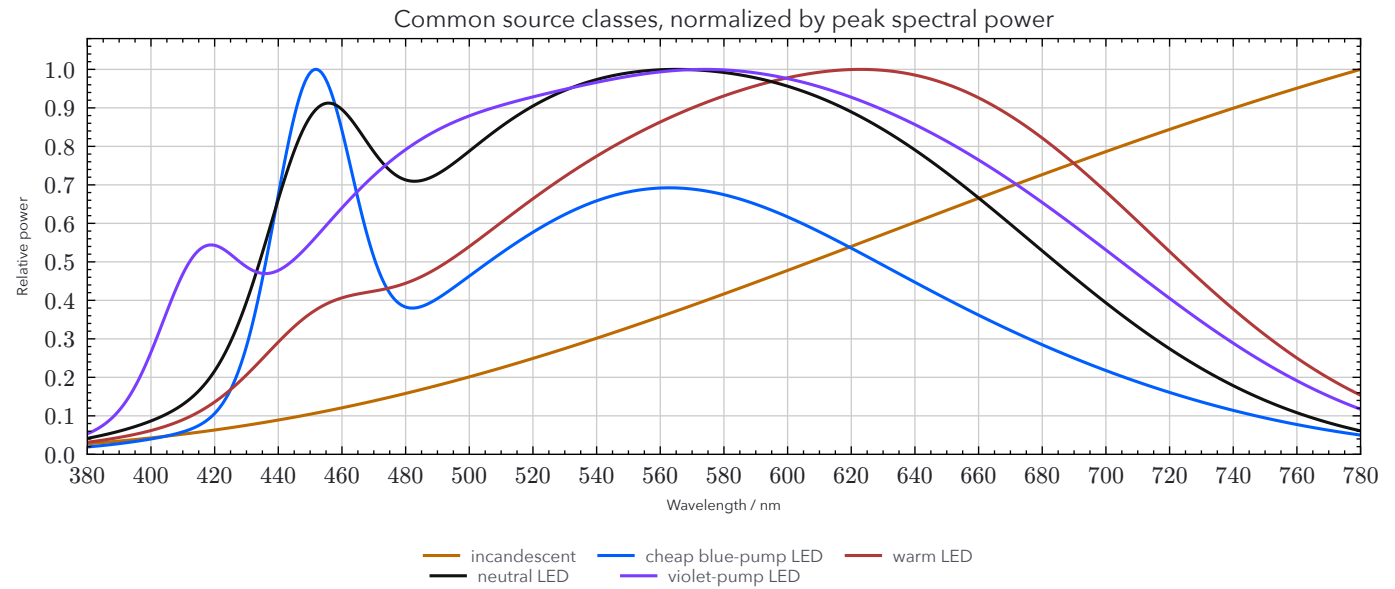
DESIGN CONSEQUENCE, FILL: BLUE

Equal white is not equal signal

Two sources can share CCT, CRI, or visual brightness and still produce different circadian signal.

White light has different engines.

The source matters. Consumer lamps that all look “white” can have radically different spectral construction.



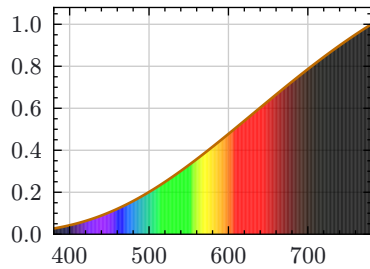
INCANDESCENT 2700 K

CHEAP BLUE-PUMP LED

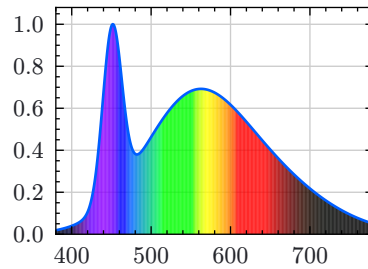
NEUTRAL PHOSPHOR LED

WARM PHOSPHOR LED

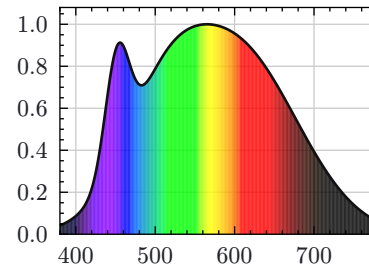
VIOLET-PUMP / FULL-SPECTRUM LED



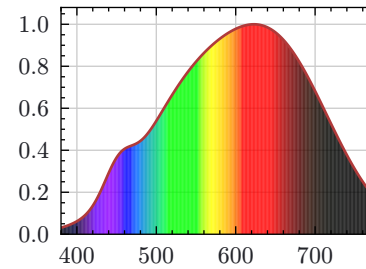
continuous, warm-heavy



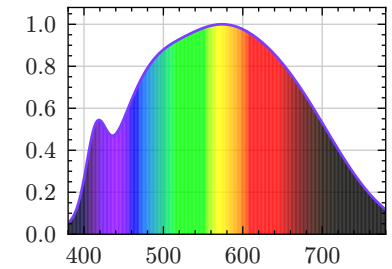
narrow blue spike + phosphor



common fixed white



blue reduced, not erased



broader engineered white

This page is the comparison anchor. The argument is not “LED bad” or “warm good.” It is: inspect the actual spectral engine before making claims about visual quality or biological signal.

source comparison

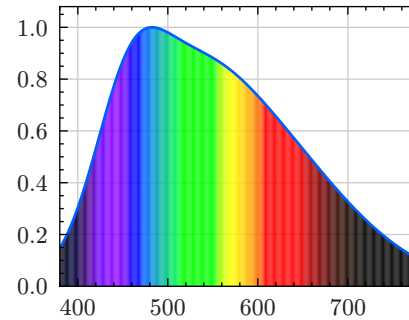
Good lighting is a sequence, not a setting.

The idealized system changes spectral job across the day: anchor, soften, protect.

This is the cleanest way to explain the design target. Do not ask one static spectrum to behave like morning daylight, evening comfort light, and biological night protection.

STATE 01

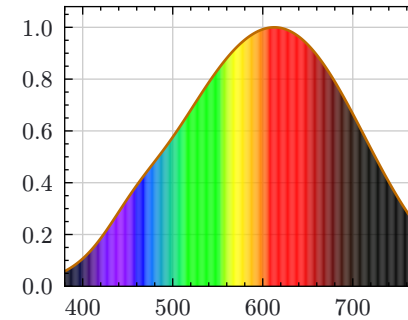
Day anchor



VISUAL high visual output with
CLOCK broad white appearance
intentional blue-cyan
presence

STATE 02

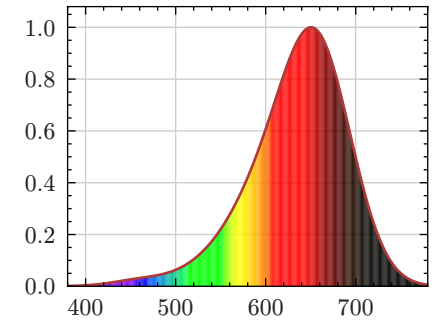
Evening transition



VISUAL comfortable, warm, still
CLOCK usable
reduced clock signal

STATE 03

Night protection



VISUAL very low task-light role
CLOCK melanopic channel
minimized

ANCHOR, FILL: BLUE

Morning / daytime

The system is allowed to be biologically loud when the day calls for anchoring and alertness.

SOFTEN, FILL: AMBER

Late day

The system separates visual comfort from biological activation instead of merely dimming the same spectrum.

PROTECT, FILL: RED

Biological night

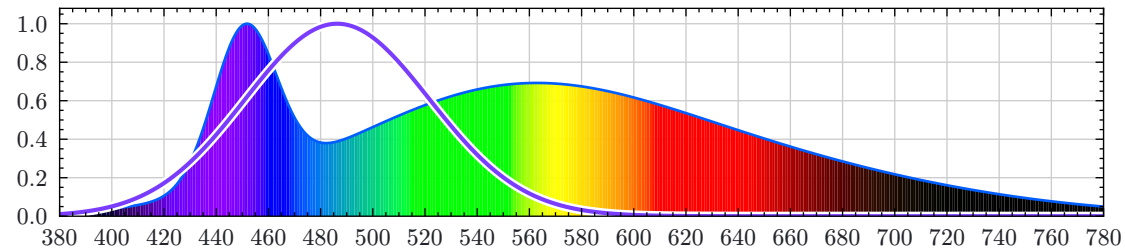
The design target becomes absence of blue-cyan signal, not a prettier warm-white marketing label.

The clock sees overlap.

Melanopic impact is not a vibe. It is the overlap between the source spectrum and the blue-cyan sensitivity region.

The violet curve is the same on every card. What changes is how much each consumer source feeds that region when compared at equal visual output.

CHEAP BLUE-PUMP LED

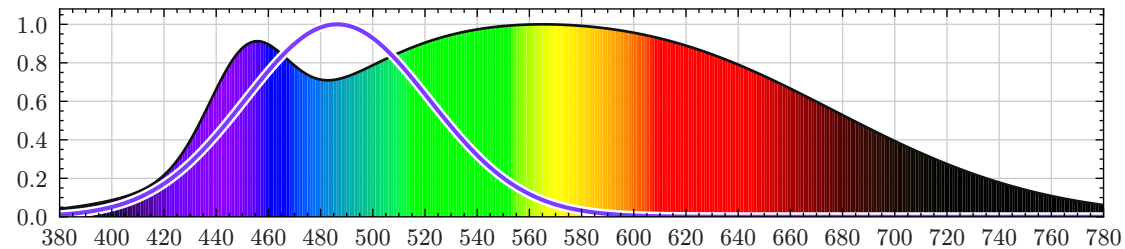


MELANOPIC OVERLAP, FILL: VIOLET

100%

relative to the highest source here, normalized to equal visual output

NEUTRAL PHOSPHOR LED

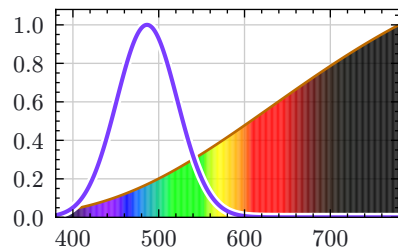


MELANOPIC OVERLAP, FILL: VIOLET

94%

relative to the highest source here, normalized to equal visual output

INCANDESCENT 2700 K

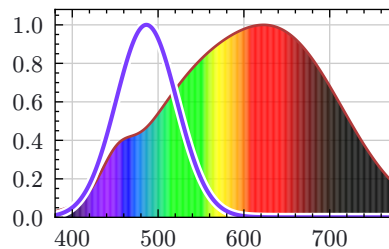


MELANOPIC OVERLAP, FILL: VIOLET

55%

relative to the highest source here, normalized to equal visual output

WARM PHOSPHOR LED

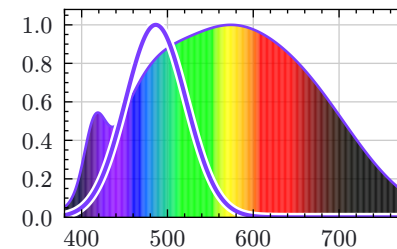


MELANOPIC OVERLAP, FILL: VIOLET

68%

relative to the highest source here, normalized to equal visual output

VIOLET-PUMP / FULL-SPECTRUM LED



MELANOPIC OVERLAP, FILL: VIOLET

92%

relative to the highest source here, normalized to equal visual output

This is the page that drives the practical point home: the consumer comparison must be made spectrally. "Dimmer," "warmer," and "white" do not describe the clock signal by themselves.

melanopic overlap